MIL-G-18694A(OS)
2 June 1975
SUPERSEDING
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14 June 1955

MILITARY SPECIFICATION

GLYCEROL - WATER MIXTURES

(FOR HYDRAULIC BUFFERS AND HYDRAULIC RECOIL MECHANISMS)

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for glycerol-water mixtures for use in hydraulic buffers and hydraulic recoil mechanisms. The methods described in this specification shall also be used for the preparation of such mixtures by the using activities.

1.2 Classification. Glycerol-water mixtures shall be of the following types, as specified in the contract or order (see 6.2):

Type A . . . 80-20 (glycerol-water) Mixture without additives.
Type B . . . 60-40 (glycerol-water) Mixture with additives.

2. APPLICABLE DOCUMENTS

2.1 Referenced documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

O-G-491	Glycerol, Technical (High Gravity)
O-S-571	Sodium Carbonate, Anhydrous, Technical
O-S-588	Sodium Chromate, Anhydrous, Technical
O-S-642	Sodium Phosphate, Tribasic, Anhydrous; Dodecahydrate; and Monohydrate; Technical

FSC 6810

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PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-C-569	Containers, Plastic, Molded (for Liquids, Pastes and Powders): Overpacked
PPP-C-1337	Containers, Metal, with Polyethylene Inserts

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

STANDARDS

American Society for Testing and Materials (ASTM) Standard

D 1193-70	Standard Specification for Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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CLASSIFICATION RULES

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. The materials used shall be such as to produce Type A and Type B mixtures in compliance with the requirements of this specification. The quality of the materials used in each mixture shall be so controlled as to insure uniformity of performance of each mixture.

3.1.1 Type A mixture. The materials used in the Type A mixture shall be as follows:

Glycerol . . . This material shall be in accordance with the requirements of O-G-491 for a single grade of glycerol known as high-gravity glycerin.

Water . . . . This material shall comply with the requirements of ASTM D 1193-70, Type I reagent water.

3.1.2 Type B mixture. The materials used in the Type B mixture shall be in accordance with the specifications and standard listed in Table I.

Table I. — Materials Used in Type B Mixture

MATERIAL	SPECIFICATIONS AND STANDARD
Glycerol	O-G-491; single grade known as high-gravity glycerin
Water	ASTM D 1193-70; Type I reagent water
Sodium carbonate	O-S-571; Type I, ordinary or light ash
Sodium chromate	O-S-588; one type and grade
Sodium phosphate	O-S-642; Type I, anhydrous

Note: Unless otherwise indicated, references to water in this specification shall be understood to mean Type I reagent water as specified in ASTM D 1193-70.

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3.2 Chemical composition.

3.2.1 Type A mixture. The chemical composition of the Type A mixture shall conform to the following requirements:

Material	Percent by Volume	Percent by Weight
Glycerol	80.0	83.4
Water	20.0	16.6

The Type A mixture shall be prepared by adding, gradually and with constant stirring, 80 parts by volume glycerol to 20 parts by volume water until thoroughly mixed.

3.2.2 Type B mixture. The chemical composition of the Type B mixture shall conform to the following requirements:

Material	Percent by Volume	Percent by Weight
Glycerol	59.6	63.9
Water	39.6	33.7
Sodium carbonate, Sodium chromate, Sodium phosphate	0.8	2.4

The Type B mixture shall be prepared as follows:

a. To 50.65 ounces of water add the following:
2.40 ounces (*avdp) of sodium carbonate,
0.14 ounce (avdp) of sodium chromate, and
0.40 ounce (avdp) of sodium phosphate.

b. Make certain that the sodium salts have entirely dissolved in the water, then add 76.25 ounces of glycerol gradually to the solution with constant stirring until thoroughly mixed.

3.3 Separation. Type A and Type B mixtures shall be homogeneous single-phase mixtures. There shall be no separation of materials and the mixtures shall be free from layers, particulate matter, precipitated sediment, and cloudiness.

3.4 Alkalinity. Type B mixture shall show an alkaline reaction when tested as specified in 4.4.4.

* avdp . . . avoirdupois

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3.5 Specific gravity. The specific gravities of Type A and Type B mixtures shall conform to the following requirements:

Mixture	Specific Gravity
Type A (high-gravity type)	1.206 ± 0.002 at 15.6°C (60°F)
Type B (low-gravity type)	1.173 ± 0.002 at 15.6°C (60°F)

3.6 Color. Type A and Type B mixtures shall be uniform in color. The Type A mixture shall be colorless. The Type B mixture shall be yellowish-green.

3.7 Workmanship. The workmanship shall be in accordance with commercial practices covering the preparation and shipment of Type A and Type B mixtures. Care shall be exercised to assure thorough mixing and correct proportions deemed necessary to provide mixtures which conform to prescribed requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lotting. A lot shall consist of a mixture, of one type, prepared by one supplier, at one plant, from the same materials, and submitted for inspection at one time.

4.3 Sampling.

4.3.1 Examination of preparation for delivery. The sample unit for this inspection shall be one shipping container, fully packed, selected just prior to the shipping container closing operation.

4.3.2 Examination of containers. A random sample of containers, filled with one type of mixture, shall be selected in accordance with General Inspection Level II of MIL-STD-105. The Acceptable Quality Level shall be 2.5 percent defective.

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4.3.3 Samples for testing. Samples for testing shall consist of a minimum of one quart of the same type of mixture which shall be made up of quantities of 8 ounces or less taken from a sufficient number of randomly selected containers and placed in separate, clean, dry, glass containers. Each sample shall be thoroughly mixed and each glass container shall be prominently identified with a durable tag, securely attached and marked with the following information:

a. Sample for Test
b. Glycerol-Water Mixture for Hydraulic Buffers and Hydraulic Recoil Mechanisms
c. Type of Mixture
d. Applicable Specification
e. Name of Manufacturer, Address and Plant (where prepared)
f. Date of Manufacture
g. Sample Submitted By (name and date)

4.4 Quality conformance inspection. Sample filled containers, selected in accordance with 4.3.2, shall conform to the stated acceptability criteria of MIL-STD-105. Rejected samples may be resubmitted for the examination, specified in 4.4.2, in accordance with the provisions of MIL-STD-105. If any sample, selected in accordance with 4.3.3, fails to conform to the requirements of 3.3, 3.4, 3.5, and 3.6, the lot represented by the sample shall be rejected.

4.4.1 Examination of preparation for delivery. A visual inspection shall be made to ensure that packaging, packing, quantity and markings comply with the reuirements of Section 5. Shipping containers, fully prepared for delivery also shall be inspected for conformance to the requirements of Section 5.

4.4.2 Examination of containers. Filled containers, selected in accordance with 4.3.2, shall be visually inspected for leakage, damage, and marking. Closures shall be inspected to ensure that each is closed tightly so as to prevent leakage of content.

4.4.3 Examination of mixtures for separation and color. Samples, selected in accordance with 4.3.3, shall be visually inspected for separation (see 3.3) and for color (see 3.6). The development of a slight cloudiness or turbidity shall not be cause for rejection.

4.4.4 Test methods for alkalinity. The Type B mixture shall be tested for alkalinity using a 10 percent phenolphthalein indicator solution. The Type B mixture shall be considered to have the required alkalinity if the color of a small sample of the mixture (selected in accordance with 4.3.3), mixed with an equal volume of water, changes to pink or red when a few drops of the indicator solution are added to the sample.

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Note: If Type B mixture under preparation by a using activity fails to show an alkaline reaction (does not change color) when tested, the alkalinity shall be adjusted as follows:

a. Prepare a solution made up of 2.5 ounces (avdp) of sodium carbonate and 0.5 ounces (avdp) of sodium phosphate dissolved in a pint of warm water.

b. Add small portions of the solution to the Type B mixture with constant stirring. After each addition, draw off a small test sample of mixture. The mixture has the required alkalinity when a few drops of indicator solution added to the sample causes it to turn pink.

4.4.5 Test methods for specific gravity. Samples of Type A and Type B mixtures, selected in accordance with 4.3.3, shall be tested at 15.6°C (60°F) for compliance with the requirements of 3.5. Determination of specific gravity shall be made with a specific-gravity hydrometer graduated in divisions of 0.002. Range of hydrometer shall depend upon the type of mixture to be tested.

Mixture	Range of Hydrometer for Test
Type A (high-gravity type)	1.200 to 1.400
Type B (low-gravity type)	1.000 to 1.200

Note: When the specific gravity of a mixture under preparation by a using activity fails to conform to the requirements of 3.5, the specific gravity shall be adjusted as follows:

a. If the specific gravity is low, add glycerol in small portions to the mixture with constant stirring until the required specific gravity is attained.

b. If the specific gravity is high, add water in small portions to the mixture with constant stirring until the required specific gravity is attained.

5. PREPARATION FOR DELIVERY

Note: Type A and Type B mixtures shall be supplied in one-quart, one-gallon or five-gallon quantities.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-quart quantity. This quantity shall be packaged as specified in PPP-C-569 for a Class A molded plastic container in a Type I fiberboard box.

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5.1.1.2 One-gallon quantity. This quantity shall be packaged as specified in PPP-C-569 for a Class B molded plastic container in a Type I fiberboard box.

5.1.1.3 Five-gallon quantity. This quantity shall be packaged as specified in PPP-C-1337 in a Type II, Class 1 metal container with polyethylene insert.

5.1.2 Level C. One-quart, one-gallon, and five-gallon quantities shall be packaged in accordance with the manufacturer's/supplier's commercial practice.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-quart quantities. Twelve, one-quart containers, packaged as specified in 5.1.1.1, shall be packed in snug fitting boxes conforming to PPP-B-621, Class 2; PPP-B-601, Overseas Type; or PPP-B-636, Type CF, Class weather-resistant.

5.2.1.2 One-gallon quantities. Six, one-gallon containers, packaged as specified in 5.1.1.2, shall be packed in a box conforming to PPP-B-636, Type CF, Class-weather resistant, Variety SW, Grade V3c.

5.2.1.3 Five-gallon quantity. One, five-gallon container, packaged as specified in 5.1.1.3, shall require no further packing.

5.2.2 Level C. Quantities, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at the lowest rate to the point of destination. Containers and packing shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Each container shall be marked in accordance with MIL-STD-129 and with any additional marking as specified by the procuring activity (see 6.2). Individual containers of Type A and Type B mixture also shall be marked as follows:

CONTAINER	MARKING ON CONTAINER
Type A mixture	Do Not Use When Temperature is Below -17.8°C (0°F)
Type B mixture	Do Not Use When Temperature is Below -28.9°C (-20°F)

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In addition, each container shall be marked with the following information:

CAUTION
Protect from freezing and avoid contamination of mixture by storing only in closed container. Freezing will not damage mixture but may damage container. Contaminated mixture may damage mechanism in which used.

6. NOTES

6.1 Intended use. Glycerol-water mixtures are generally used in hydraulic type buffers and hydraulic (throttling) type recoil mechanisms of enclosed gun and mount assemblies, turrets, and other enclosed equipment. The type of mixture (Type A or Type B) used in any particular equipment shall be that specified in instructions or drawings for the equipment.

a. Type A mixture is used in equipment where temperatures lower than -17.8°C (0°F) are not normally encountered.

b. Type B mixture is used in equipment where temperatures of -17.8°C to -28.9°C (0°F to -20°F) may be encountered.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of mixture required (see 1.2 and 6.1).
c. Quantity required (see Section 5).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Special markings required (see 5.3).

Note: This specification has been extensively revised. Therefore, the margins of this specification have not been marked with asterisks to indicate where changes from the previous issue were made.

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DOCUMENT IDENTIFIER AND TITLE
MIL-G-18694A(OS), "Glycerol-Water Mixtures"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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